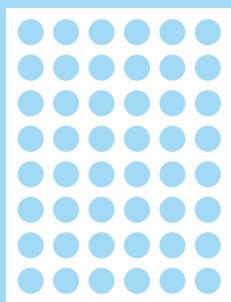


Certificate of Analysis



Olink[®] Explore

PROJECT NAME

ISSUE DATE

9/9/2022

1. Project information

No. of samples	No. of plates	Normalization method
920	11	Intensity normalization

1.1 Sample type

N/A

1.2 Project specific comments

The following Explore 384 Oncology assays did not meet Olink's batch release quality control criteria and are therefore not included in this project: SMAD1 and ARHGAP25.

2. Quality control

Three internal controls are added to each sample, the Incubation control, the Extension Control and the Amplification control. The Extension Control is used for the generation of the NPX values. The Incubation Control and the Amplification Control are used to monitor the quality of assay performance, as well as the quality of individual samples.

Three external controls are included in each run, the Plate Control (healthy pooled plasma), Sample Control (healthy pooled plasma) and Negative Control. The Plate Control is used for data normalization, the Sample Control is used to assess potential variation between runs and plates, and the Negative Control is used to calculate Limit of Detection for each assay and to assess potential contamination of assays.

The following parameters are evaluated in the Quality Control(QC) :

1. The average matched counts¹ for each sample. To pass QC, there should be at least 500 counts, otherwise the sample receives a QC warning status.
2. The deviation from the median value of the Incubation- and Amplification Controls for each individual sample. To pass QC, the deviation should not exceed +/-0.3 NPX for either of the internal controls, otherwise the sample will receive a QC warning status.
3. The deviation of the median value of the Negative Controls from a predefined value set for each assay. To pass QC, the deviation of the median of the Negative Controls must be ≤ 5 standard deviations from the set predefined value, otherwise the assay will receive a warning status.

¹The number of reads for each specific combination of sample and assay

All samples included in the project are presented in the data output file. Samples that do not pass the QC are indicated with WARN in the column named QC_warning. Data points from samples that do not pass QC should be treated with caution. Manual QC warnings are indicated with MANUAL_WARN in the column named QC_warning. Section 2.1 reports the fraction of samples that pass QC for all assays per panel and the fraction of data points passing QC per panel. Samples with manual QC warning are counted as not passed QC. Assays that do not pass the QC are indicated with WARN in the column named Assay_warning. Data points from assays that do not pass QC should be treated with caution.

2.1 QC summary

Olink Panel	Samples passed QC	Samples passed QC (%)	Datapoints passed QC	Datapoints passed QC (%)
Explore 384 Cardiometabolic	830 / 920	90	330295 / 339480	97
Explore 384 Inflammation	838 / 920	91	330315 / 338560	98
Explore 384 Neurology	807 / 920	88	322719 / 337640	96
Explore 384 Oncology	774 / 920	84	316156 / 336720	94

2.2 Intra - and Inter - assay Coefficient of Variance(% CV)

Intra - and inter - CVs are based on the Sample Controls(pooled plasma samples) included on each sample plate. Calculations are made for each assay using NPX - values.Average % CV for all assays on a panel is presented in section 2.2.1.The number of assays with CVs within defined intervals are presented in section 2.2.2.

2.2.1 Average %CV

Olink Panel	Intra-assay %CV	Inter-assay %CV
Explore 384 Cardiometabolic	9	18
Explore 384 Inflammation	8	19
Explore 384 Neurology	10	22
Explore 384 Oncology	9	20

2.2.2 Intra-assay %CV distribution

Olink Panel	<5%	≥5-<10%	≥10-<15%	≥15%	N/A*
Explore 384 Cardiometabolic	21	228	79	28	13
Explore 384 Inflammation	126	136	40	43	23
Explore 384 Neurology	53	178	63	57	16
Explore 384 Oncology	111	146	43	54	12

*Assays where CV is not possible to calculate

2.2.3 Inter-assay %CV distribution

Olink Panel	<10%	≥10-<20%	≥20-<30%	≥30%	N/A*
Explore 384 Cardiometabolic	26	244	56	34	9
Explore 384 Inflammation	97	134	59	63	15
Explore 384 Neurology	10	191	89	66	11
Explore 384 Oncology	46	177	71	64	8

*Assays where CV is not possible to calculate

3. Protein detection results

3.1 Number of proteins detected in >50% of the samples

Olink Panel	No. of detected proteins / Total no. of proteins	Detected proteins (%)	Expected detectability in EDTA plasma* (%)
Explore 384 Cardiometabolic	353 / 369	96	N/A
Explore 384 Inflammation	346 / 368	94	N/A
Explore 384 Neurology	344 / 367	94	N/A
Explore 384 Oncology	350 / 366	96	N/A

*The expected detectability is based on EDTA plasma from healthy donors. These values are intended as guidelines only and protein levels may vary depending on different pathological conditions, sample matrices, or sample preparation methods.

3.2 Data output

Data is presented as NPX (Normalized Protein eXpression) values. NPX is Olink's relative protein quantification unit on log2 scale. NPX values are calculated from the number of matched counts, using NGS (Next Generation Sequencing) as readout. Data values for measurements below limit of detection (LOD) are reported for all samples.